

Applied Algebra

Spring 2026

Instructor: Tuan Tran

Exercise Sheet 3

Deadline: by the end of next Tuesday

Instructions. You are encouraged to work in pairs. When submitting your solutions, list the team members and circle the name of the person responsible for writing up the solutions that week. Each week one person should write the solutions, and each student should do so for at least five weeks during the semester.

Please submit written solutions to **TWO** of the following problems.

Problem 1. Find

- (i) $5^{20} \bmod 7$,
- (ii) $2^{16} \bmod 8$,
- (iii) $7^{1001} \bmod 11$, and
- (iv) $6^{10} \bmod 15$.

Problem 2. Two players, Alice and Bob, start with the number 1. They take turns multiplying the current number by 2, 3, or 5, reducing modulo 41 after each move. The player who first makes the number equal to 1 again (after the starting position) wins.

Determine which player has a winning strategy, and justify your answer.

Problem 3. Show that if $n \geq 2$ and p is a prime dividing n such that $p^2 \nmid n$, then

$$p^{\varphi(n)+1} \equiv p \pmod{n}.$$

Problem 4. A word has been broken into blocks of two letters and converted to two-digit numbers using the correspondence

$$a = 0, b = 1, c = 2, d = 3, o = 4, k = 5, f = 6, h = 7, l = 8, j = 9.$$

The blocks are then encoded using the public key code $(87, 19)$. The encrypted message is 04/10. Find the original message.